StudyFyX - An Integrated Explainable AI Platform for Academic Guidance, Careers, and Wellbeing

Panel Comment for proposal	**How our team addressed it**
The panel noted a lack of proper methodology for gathering accurate project datasets.	We implemented custom web scrapers, user surveys, and mentor interviews to collect and validate data.

StudyFyX - An Integrated Explainable AI Platform for Academic Guidance, Careers, and Wellbeing

- ## Problem & Its Impact / Gravity

- Students struggle to choose the correct degree and university after A/L examinations.
- Many students lack proper career guidance and awareness about future job opportunities.
- There is a gap between university education and industry-required skills.
- Students face academic stress and poor study management without personalized learning support.
- Online mentorship and career advice platforms are often unverified and unreliable.
- Wrong career and degree decisions can lead to unemployment, low job satisfaction, and skill mismatch.

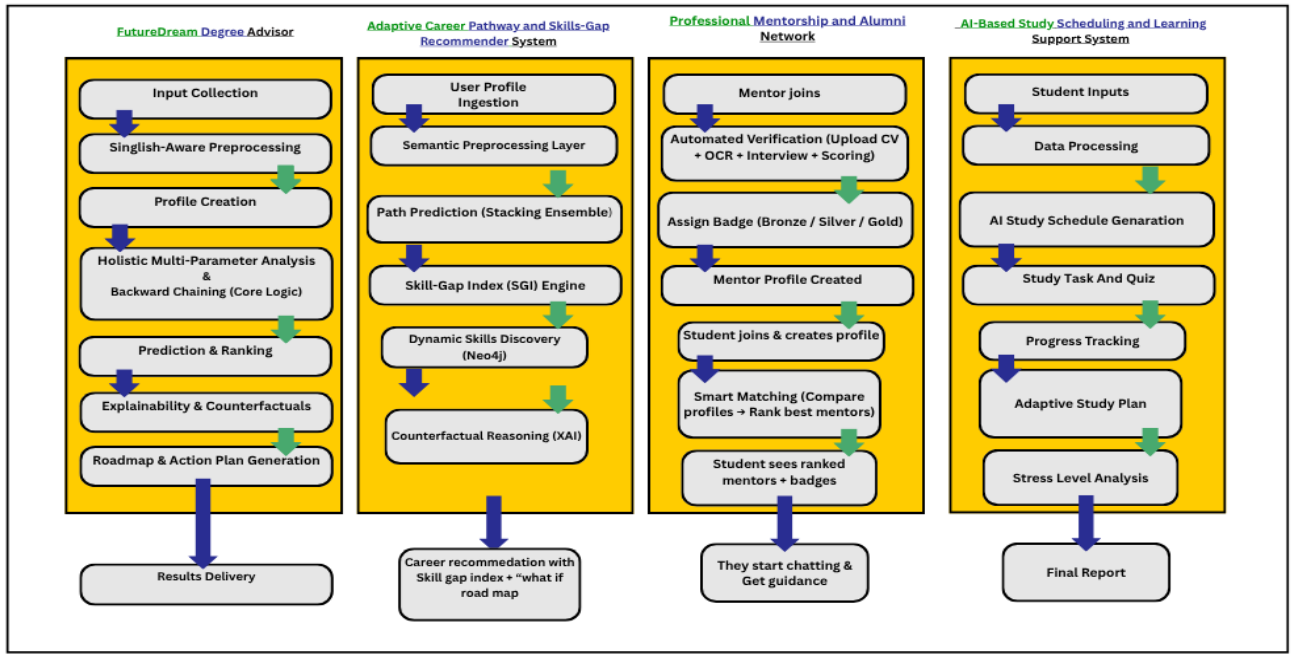
- ## Design Excellence of the Overall Solution

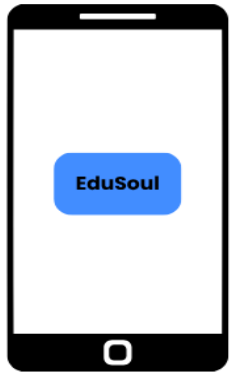
- An intelligent AI-powered platform that combines career guidance, skill-gap analysis, verified mentorship, and adaptive study planning into one personalized system for Sri Lankan students.

StudyFyX






FutureDream Degree Advisor
Input Collection
Singlish-Aware Preprocessing
Profile Creation
Holistic Multi-Parameter Analysis & Backward Chaining (Core Logic)
Prediction & Ranking
Explainability & Counterfactuals
Roadmap & Action Plan Generation
Results Delivery
Adaptive Career Pathway and Skills-Gap Recommender System
User Profile Ingestion
Semantic Preprocessing Layer
Path Prediction (Stacking Ensemble)
Skill-Gap Index (SGI) Engine
Dynamic Skills Discovery (Neo4j)
Counterfactual Reasoning (XAI)
Career recommedation with Skill gap index + "what if road map
Professional Mentorship and Alumni Network
Mentor joins
Automated Verification (Upload CV + OCR + Interview + Scoring)
Assign Badge (Bronze / Silver / Gold)
Mentor Profile Created
Student joins & creates profile
Smart Matching (Compare profiles → Rank best mentors)
Student sees ranked mentors + badges
They start chatting & Get guidance
AI-Based Study Scheduling and Learning Support System
Student Inputs
Data Processing
AI Study Schedule Genaration
Study Task And Quiz
Progress Tracking
Adaptive Study Plan
Stress Level Analysis
Final Report


EduSoul



FutureDream Degree Advisor

IT22220288 | RUSIRU L.W.S

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FUTUREDREAM DEGREE ADVISOR

- How the Solution Addresses the Sub-Problem

• Many students do not know which degree and university are most suitable for their future career goals.

• This system starts from the student's dream job and analyses academic results, personality, interests, and lifestyle preferences to recommend the best degree pathway and suitable university options.

• It also provides a long-term roadmap to help students achieve their future goals.

- User Requirements Addressed

- University and Degree recommendations
- Backward dream goal analysis and Personalized roadmap
- Personality analysis - Future Works
- Multilingual support - Future Works

FUTUREDREAM DEGREE ADVISOR

• Design Excellence / Contribution

- First backward career guidance system for Sri Lankan students - Backward Chaining Model
- Combines academic, personality, and lifestyle factors
- Explainable AI recommendations - EnhancedXAI
- Personalized long-term roadmap

• **User Feedback on Prototype**

•Positive Feedback

•Easy to understand

•Helpful career recommendations

•Useful roadmap guidance

• Improvements Suggested

- Combines academic, personality, and lifestyle factors
- Improve mobile UI



ADAPTIVE CAREER PATHWAY AND SKILL-GAP ANALYZER

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ADAPTIVE CAREER PATHWAY AND SKILL-GAP ANALYZER

How the Solution will address the Sub-Problem / Prototype

- Analyzes local job market trends via web scraping.
- Predicts top career matches using ML models.
- Calculates a personalized "Skill-Gap" roadmap.

User Requirements addressed by the solution

- Smart, real-time skill searching.
- Context-aware career matching.
- Securely saving user progress.

ADAPTIVE CAREER PATHWAY AND SKILL-GAP ANALYZER

Design Excellence/Contribution

- Clean, modern mobile user interface.
- Lightning-fast user interactions.
- Engineered a Custom "Affinity Boost" algorithm.

User Feedback on Prototype

- Skill input was very convenient.
- High clarity on learning roadmaps.
- The "Readiness Score" was motivating.



Professional Mentorship and Alumni Network

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PROFESSIONAL MENTORSHIP AND ALUMNI NETWORK

- How the Solution Addresses the Sub-Problem

• Helps students find trusted and qualified mentors based on academic interests and career goals.
• Uses AI-based mentor–mentee matching to recommend suitable mentors.
• Verifies mentor credibility using CV and certificate uploads with OCR-based information extraction.
• Conducts structured interview scoring to evaluate mentor authenticity and communication skills.
• Provides secure real-time communication between students and mentors.
• Protects user privacy through secure document handling .
• Supports students in career planning and professional development.

- User Requirements Addressed

- Secure mentor verification system.
- AI-powered mentor interview assessment .
- Professional CV analysis and scoring.
- Avatar for live interview experience - Future Works
- Implementing a secure chat platform - Future Works
- Ai student matching implement for mobile app - Future Works

PROFESSIONAL MENTORSHIP AND ALUMNI NETWORK

• Design Excellence / Contribution

• Modern and user-friendly UI/UX

• Responsive cross-platform design

• Professional dashboard experience

• Interactive and engaging interface

• User Feedback on Prototype

• Positive Feedback

• Easy to use
• Clear recommendations
• Helpful guidance
• Smooth user experience

• Improvements Suggested

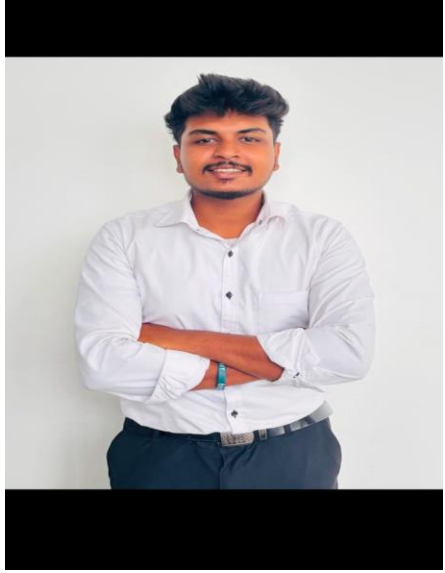
• Add more career options

• Improve recommendation accuracy

• Enhance mobile experience

• Include real-time chat support

Smart Study Planning | Real-time Stress Management And Mental Wellbeing



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SMART STUDY PLANNING | REAL-TIME STRESS MANAGEMENT AND MENTAL WELLBEING

- How the Solution Addresses the Sub-Problem

- Helps university students effectively manage heavy academic workload and prevents burnout through intelligent time management.
- Analyzes student's timetable, assignments, exams, working hours, and smartwatch data to calculate accurate workload score.
- Automatically generates personalized daily study plans based on available time slots and priority.
- Regular updates with new Sri Lankan university academic calendars, exam patterns, and stress management techniques.

- User Requirements Addressed

- Real-time stress monitoring with Low, Moderate, and High levels.
- Weekly performance report with insights and recommendations.
- Personalized suggestions and motivational support based on stress and performance.

SMART STUDY PLANNING | REAL-TIME STRESS MANAGEMENT AND MENTAL WELLBEING

• Design Excellence / Contribution

- First intelligent workload and mental wellbeing assistant specifically designed for Sri Lankan university students.
- Integrates academic data (timetable, assignments, exams) with lifestyle data (sleep, working hours) and physiological data (smartwatch).
- Delivers personalized daily study plans and weekly performance reports with motivational guidance.

•User Feedback

•Positive Feedback

- Easy to understand
- Useful weekly reports that show clear progress and weak areas.
- Helps reduce academic overwhelm and improves time management.

• Improvements Suggested

- Need for more attractive and modern mobile UI/UX design.
- Include integration with actual smartwatch APIs

StudyFyX

Commercialization / Sustainability of Solution

- AI-powered educational and career guidance platform for students
- Can be provided to schools, universities, and educational institutes
- Supports long-term sustainability through yearly dataset and job market updates
- Can be expanded as a web and mobile platform across Sri Lanka

Market Size

•Thousands of Sri Lankan students sit for A/L examinations annually

•Large demand for career guidance, mentorship, and academic support systems

•Increasing interest in AI-based educational technologies in Sri Lanka

StudyFyX

What is Unique in Your Solution

- Combines AI career guidance, mentorship, and study assistance in one platform
- Uses backward career analysis from dream job to degree pathway
- Provides Skill-Gap analysis and personalized career roadmaps
- Includes verified mentor support with trusted badges
- Adaptive AI study planner with stress and workload analysis
- Explainable AI (XAI) and counterfactual recommendations
- Sri Lanka-specific university and Z-score analysis
- Trilingual and Singlish-aware user interaction

How Will the Cost Be Recovered?

- Student subscription packages
- School and university licensing
- Premium mentorship and counseling services
- Advertisement and educational partnerships
- Career assessment and premium AI guidance features

Undergraduate Student Persona



Sarith Samarakoon

Software Engineer

Smart | Honest | Loyal

BIO

Undergraduate Software Engineering student passionate about mobile app development, AI, and building innovative software solutions.

GOALS

- Manage study schedules effectively
- Reduce academic stress
- Improve learning performance

SKILLS

Problem Solving

Public Speaking

MOTIVATION

- Poor time management
- Stress during assignments and exams
- Difficulty maintaining study consistency

PERSONALITY

Design Thinking

Resource

FAVORITE BRAND